

Recipe Analytics Pipeline Report

ETL Analytics Module

November 24, 2025

1 Overview

This report summarizes the insights derived from the ETL pipeline analytics module. The data analyzes recipe ingredients, complexity, engagement metrics, and author contributions.

2 Ingredient Analysis

The dataset reveals a preference for staple ingredients. Rice and Garlic are the most frequently used items across all recipes.

Table 1: Top 5 Most Used Ingredients

Ingredient	Count
Rice	21
Garlic	19
Onion	16
Salt	15
Oil	13

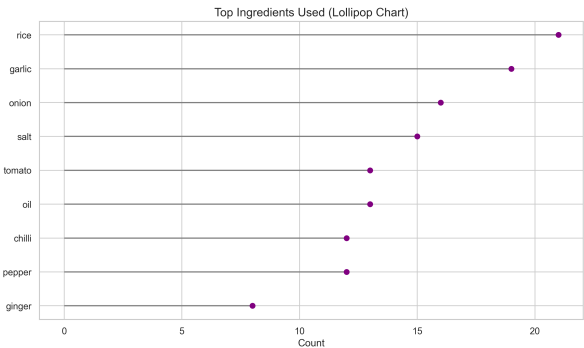


Figure 1: Top most uswed Ingredients

3 Recipe Complexity & Cuisine

3.1 Preparation Time and Difficulty

The **average preparation time** across the dataset is **12.43 minutes**. The difficulty distribution is balanced between Easy and Medium levels.

Table 2: Recipe Difficulty Distribution

Difficulty Level	Count
Easy	11
Medium	11
Hard	7
Invalid Level	1

3.2 Cuisine Analysis

We observed distinct difficulty patterns across cuisines. For example, South Indian recipes show a higher percentage of "Hard" difficulty ratings (37.50%) compared to Punjabi recipes (14.29%).

Table 3: Average Rating per Cuisine

Cuisine	Avg Rating
Punjabi	4.36
Hyderabadi	4.14
South Indian	3.94
Bengali	3.92

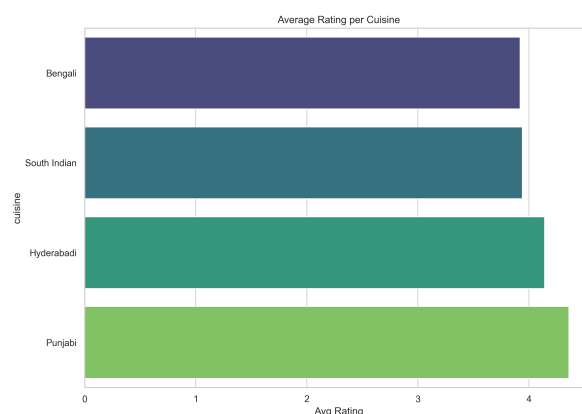


Figure 2: Average rating per cuisine

4 Engagement Metrics

4.1 Engagement vs. Difficulty

An interesting inverse correlation exists between rating and engagement volume regarding difficulty. While "Easy" recipes have higher average user ratings (4.17), "Hard" recipes generate significantly higher average engagement (712.86).

Table 4: Performance Metrics by Difficulty

Difficulty	Avg Rating	Avg Engagement
Easy	4.17	621.64
Medium	4.13	534.64
Hard	3.93	712.86

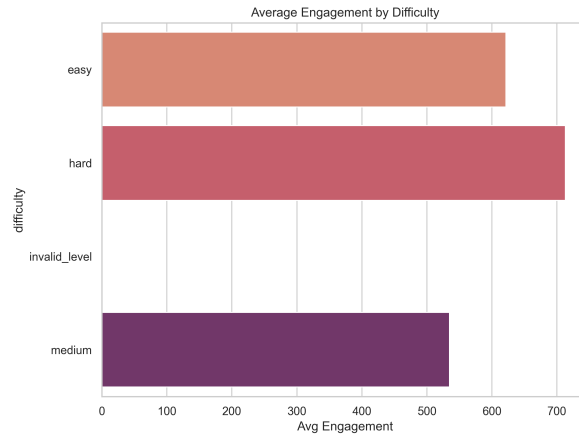


Figure 3: Engagement vs difficulty

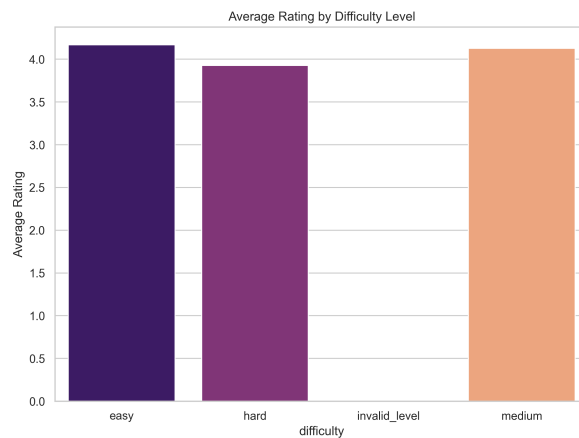


Figure 4: Average rating of recipe vs difficulty level

Correlation Note: The correlation between *Prep Time* and *Total Likes* is **-0.093**,

indicating no meaningful relationship between how long a dish takes to cook and how many likes it receives.

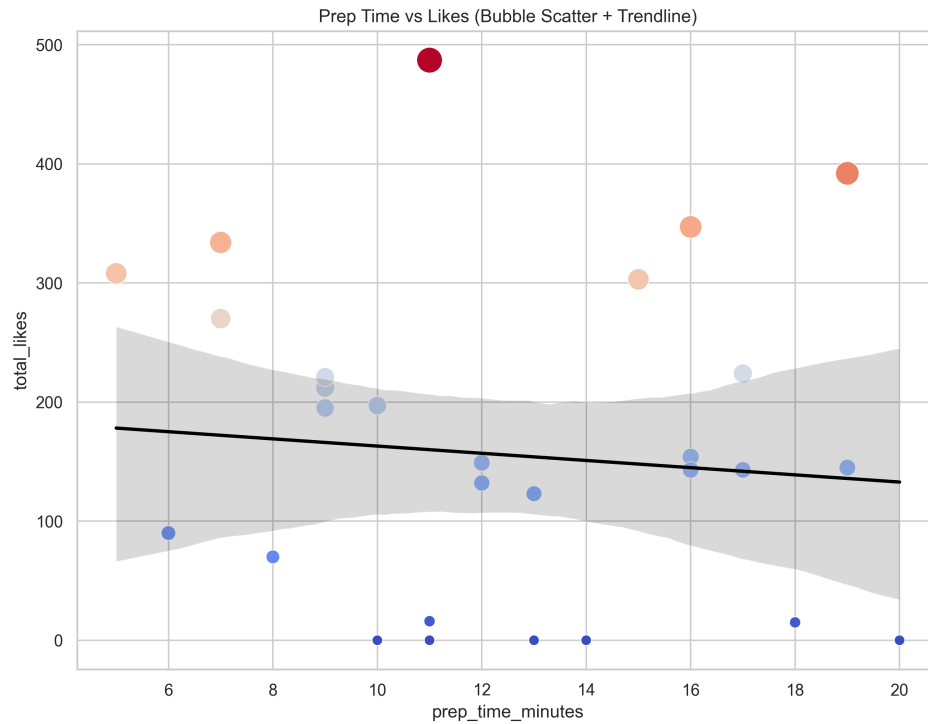


Figure 5: Preptime vs likes

5 Leaderboards

5.1 Top Performing Recipes

The table below highlights recipes with the highest total views. "Veg Kurma" is the clear outlier in popularity.

Table 5: Top 5 Viewed Recipes

Recipe ID	Title	Total Views
r_006	Veg Kurma	1354
r_011	Egg Bhurji	1136
r_013	Chicken Curry	1008
r_008	Hyderabadi Biryani	979
r_001	Pav Bhaji	922

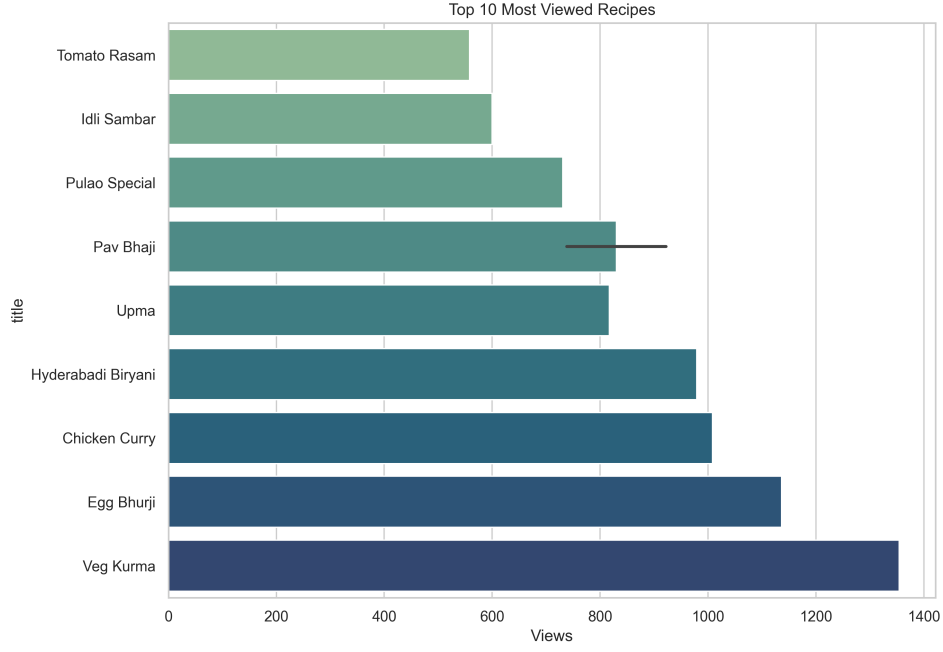


Figure 6: Top Viewed Recipes

5.2 Top Authors

User contribution indicates that **user_04** is the most prolific contributor.

Table 6: Top Content Creators

User ID	Recipes Created
user_04	6
user_03	5
user_10	5
user_07	5
user_09	4